

第9週「運動の表し方」小テスト 解答

問 1: $x-t$ 図

グラフは $(t, x) = (0, 0), (2, 4), (4, 4), (6, 0), (8, -4), (10, -4)$ を直線で結んだもの。

1. $v = (4 - 0)/(2 - 0) = 2.0 \text{ m/s}$
2. $v = (4 - 4)/(4 - 2) = 0 \text{ m/s}$
3. $v = (0 - 4)/(6 - 4) = -2.0 \text{ m/s}$
4. $v = (-4 - 0)/(8 - 6) = -2.0 \text{ m/s}$
5. $\bar{v} = (-4 - 0)/(10 - 0) = -0.40 \text{ m/s}$

問 2: $v-t$ 図

グラフは $(t, v) = (0, 0), (2, 4), (4, 4), (6, 0), (8, -4), (10, -4)$ を直線で結んだもの。

6. $a = (4 - 0)/(2 - 0) = 2.0 \text{ m/s}^2$
7. $a = (4 - 4)/(4 - 2) = 0 \text{ m/s}^2$
8. $a = (0 - 4)/(6 - 4) = -2.0 \text{ m/s}^2$
9. $\Delta x = \frac{1}{2} \times 2 \times 4 + 2 \times 4 = 12 \text{ m}$
10. $\Delta x = 4 + 8 + 4 - 4 - 8 = 4 \text{ m}$